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COURSE CODE: **SBT-DF205**
LAB NAME: **CASE 2 IPHONE SE IOS 13.4.1**
DATE: **04/11/2026**
INSTRUCTOR: **AMINU IDRIS**

In this training case, Harbour Operations Ltd requests a forensic review of a company-issued iPhone SE following a dispute about messages, shared media and the device's movements. As the examiner, establish the associated accounts and reconstruct relevant activity. Assess whether communications, browsing, photographs and device records support a consistent account of events.

Use the original identities and historical dates in the supplied dataset. Base your conclusions on the records examined and explain any findings that remain uncertain.

Evidence and tools

Use Joshua Hickman's iOS 13.4.1 evidence package hosted by Digital Corpora. Download ios_13_4_1.zip, extract it, then extract the file system archive at Extraction/13-4-1.tar inside the package. Use that file system for the graded tasks.

Download the iOS 13.4.1 evidence package

The download is approximately 9.63 GB. Allow at least 60 GB of free disk space. Work in a Linux VM or macOS to preserve Unix paths and links. Use an archive extractor, DB Browser for SQLite or SQLite, a binary/XML plist reader, ExifTool and a spreadsheet editor. Refer to the supplied iOS slides for examination procedures.

Examination instructions

- Preserve the source ZIP and TAR. Examine working copies and keep matching database -wal and -shm files together.
- Work only on the supplied extraction. Jailbreaking, resetting devices and accessing live accounts are outside this activity.
- Record source paths and record IDs. Treat /var and /private/var aliases carefully to avoid counting the same evidence twice.
- Retain raw timestamps, normalise the timeline to UTC and document empty or unavailable records. Exclude passwords, tokens and unrelated personal data from your report.

Practical investigation tasks

Task 1 Evidence preservation 10 marks

1. Create a handling log using case reference SBT-DF205-C2. Record the evidence source, archive names, sizes, receipt time, examiner and tool versions.

2. Calculate SHA-256 values for the source ZIP and TAR before examination and recheck them at completion. Compare with a trusted manifest for the exact files if supplied; otherwise record your values as baselines.

3. Maintain an evidence register for selected files, including evidence ID, source path, size and SHA-256. Record extraction errors, processing actions and database sidecars.

Task 2 Device accounts and settings 10 marks

1. Establish the device model, iOS version and build from configuration or log artefacts. Cite the exact source and property keys.

2. Examine Accounts3.sqlite and relevant plists to identify account associations. Corroborate one association using a second source.

3. Report two supported settings, such as location services, backup information or effective user settings. Distinguish stored configuration from proof of user activity.

Task 3 Messages contacts and calls 20 marks

1. Locate private/var/mobile/Library/SMS/sms.db and inspect its schema. Extract at least five messages and reconstruct one conversation, showing record IDs, participants, content, direction, service type and raw and converted timestamps.

2. Trace one message attachment record to its local file where available. Explain the relationship and any unavailable content. Validate status meanings against the schema and application version.

3. Use AddressBook.sqlitedb to associate at least two message or call identifiers with stored contacts where matches exist. Examine at least three calls in CallHistory.storedata, including direction, time and duration.

Task 4 Safari and calendar 10 marks

1. Examine Safari History.db and export at least five visits with URLs, titles, raw times and UTC conversions. Distinguish recorded visits from bookmarks, open tabs and synced tabs.

2. Inspect Calendar.sqlitedb and Extras.db where present. Record relevant tables and row counts, then report any supported event details. Document empty tables without inventing appointments.

Media activity and location analysis

Task 5 Photographs and media 15 marks

1. Examine Photos.sqlite, including ZGENERICASSET and related metadata tables as present. Select at least three media records and locate their corresponding files where available.

2. Record filenames, paths, hashes, capture-related and added times, available offsets and GPS metadata. Distinguish media captured by the camera from imported or saved media using supporting fields.

3. Validate one database result against file metadata. Explain discrepancies and any supported sharing or deletion indicators without assuming an indicator proves a complete transfer or deliberate deletion.

Task 6 KnowledgeC activity 10 marks

1. Locate private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db. Inspect ZOBJECT and related metadata, then extract at least five relevant records, such as application focus or device lock activity.

2. Report the stream name, record ID, associated application or value, raw start and end times, UTC conversions and duration where meaningful. Correlate at least one interval with messages, browsing or media activity.

Task 7 Health and location evidence 15 marks

1. Inspect healthdb.sqlite and healthdb_secure.sqlite where present. Identify a recorded data source or device and examine at least two activity samples. Report times, values and units without treating a synced wearable record as proof the phone holder performed the activity.

2. Inspect available routined Local.sqlite and locationd cache_encryptedB.db records. Examine at least three location-related entries where available, recording source tables, coordinates, times and their meanings.

3. Distinguish learned visits from Wi-Fi or cell reference locations. Use supported visit start and end fields where available; a record creation time is not automatically a visit time. Compare one location result with media or activity evidence.

Task 8 Timeline and conclusions 10 marks

1. Create a timeline of at least 12 distinct events from four or more artefact families. Include raw time, field name, epoch and unit, UTC conversion, event description and evidence ID.

2. Present at least three correlations and explain two alternative interpretations. State what the evidence establishes, what remains uncertain and what further evidence would resolve the gaps.

Time handling. Many Apple dates use a 1 January 2001 epoch, but units vary by field. Validate each conversion. The creator’s activity notes use UTC-04:00. Keep media times without a known offset explicitly unresolved.

ANSWER:


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Owasp bvwa
Metasploitable 2
Kali Linux
Home
Kali Linux
collins@kali: ~/iOS_13
Session Actions Edit View Help
(collins㉿kali)-[~/iOS_13]
$ wget https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip
--2026-11-04 21:52:16-- https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 16.15.38.179, 16.15.44.141, 52.92.165.17, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|16.15.38.179|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9629286492 (9.0G) [application/zip]
Saving to: 'ios_13_4_1.zip'
ios_13_4_1.zip 90%[++>] 8.11G --.-KB/s in 96s
2026-11-05 00:47:51 (2.47 MB/s) - Read error at byte 8704397423/9629286492 (Error in the pull function.). Retrying.
--2026-11-05 00:47:50-- (try: 4) https://digitalcorpora.s3.amazonaws.com/corpora/mobile/ios_13_4_1/ios_13_4_1.zip
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|16.15.44.141|:443... connected.
HTTP request sent, awaiting response... 206 Partial Content
Length: 9629286492 (9.0G), 924889069 (882M) remaining [application/zip]
Saving to: 'ios_13_4_1.zip'
ios_13_4_1.zip 100%[++>] 8.97G 1.91MB/s in 7m 23s
2026-11-05 00:56:18 (1.99 MB/s) - 'ios_13_4_1.zip' saved [9629286492/9629286492]
(collins㉿kali)-[~/iOS_13]
$ ls ios_13_4_1.zip -l
-rw-rw-r-- 1 collins collins 9629286492 Nov 22 2020 ios_13_4_1.zip
(collins㉿kali)-[~/iOS_13]
$ hashdeep ios_13_4_1.zip
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
Invoked from: /home/collins/iOS_13
$ hashdeep ios_13_4_1.zip
##
9629286492,4e094801ba1e3b938b2c0cc18d0e0d1d,761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551,/home/collins/iOS_13/ios_13_4_1.zip
(collins㉿kali)-[~/iOS_13]
$ unzip -q ios_13_4_1.zip
(collins㉿kali)-[~/iOS_13]
$ ls -l
total 9403612
drwxr-xr-x 6 collins collins 4096 Apr 19 2020 'iOS 13.4.1 Extraction'
-rw-rw-r-- 1 collins collins 9629286492 Nov 22 2020 ios_13_4_1.zip
(collins㉿kali)-[~/iOS_13]
$ ls -l
total 9403612
drwxr-xr-x 6 collins collins 4096 Apr 19 2020 'iOS 13.4.1 Extraction'
-rw-rw-r-- 1 collins collins 9629286492 Nov 22 2020 ios_13_4_1.zip
(collins㉿kali)-[~/iOS_13]
$ cd 'iOS 13.4.1 Extraction'
(collins㉿kali)-[~/iOS_13/iOS 13.4.1 Extraction]
$ ls -l
total 8724
drwxr-xr-x 3 collins collins 4096 Nov 5 06:40 Extraction
drwxr-xr-x 2 collins collins 4096 Apr 16 2020 'Extraction Logs'
-rw-r--r-- 1 collins collins 8916857 Apr 19 2020 iOS13-4-1-ImageCreation.pdf
drwxr-xr-x 2 collins collins 4096 Apr 16 2020 'iTunes Backup'
drwxr-xr-x 2 collins collins 4096 Apr 16 2020 'Sysdiagnose Logs'
(collins㉿kali)-[~/iOS_13/iOS 13.4.1 Extraction]
$ cd Extraction
(collins㉿kali)-[~/iOS_13/iOS 13.4.1 Extraction/Extraction]
$ tar -xvf 13-4-1.tar
Volumes/JOSH/NoTar-13-4-1/
Volumes/JOSH/NoTar-13-4-1/.ba/
Volumes/JOSH/NoTar-13-4-1/.file
Volumes/JOSH/NoTar-13-4-1/.mb/
Volumes/JOSH/NoTar-13-4-1/.Trashes
Volumes/JOSH/NoTar-13-4-1/Applications/
Volumes/JOSH/NoTar-13-4-1/bin/
Volumes/JOSH/NoTar-13-4-1/cores/
Volumes/JOSH/NoTar-13-4-1/dev/
Volumes/JOSH/NoTar-13-4-1/Developer/
Volumes/JOSH/NoTar-13-4-1/etc
Volumes/JOSH/NoTar-13-4-1/Library/
Volumes/JOSH/NoTar-13-4-1/private/
Volumes/JOSH/NoTar-13-4-1/sbin/
Volumes/JOSH/NoTar-13-4-1/System/
Volumes/JOSH/NoTar-13-4-1/tmp
Volumes/JOSH/NoTar-13-4-1/usr/
Volumes/JOSH/NoTar-13-4-1/var
Volumes/JOSH/NoTar-13-4-1/usr/bin/
Volumes/JOSH/NoTar-13-4-1/usr/include/
Volumes/JOSH/NoTar-13-4-1/usr/lib/
Volumes/JOSH/NoTar-13-4-1/usr/libexec/
Volumes/JOSH/NoTar-13-4-1/usr/local/
Volumes/JOSH/NoTar-13-4-1/usr/sbin/
Volumes/JOSH/NoTar-13-4-1/usr/share/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/Mav10.Release.bbfw
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/ramdisk/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/ramdisk/arm64SURamDisk.dmg
^C
Metasploitable 2
Kali Linux
Metasploitable 2
Kali Linux

```
┌──(collins㉿kali)-[~/…/Extraction/Volumes/JOSH/NoTar-13-4-1]
└─$ ls -l
total 40
drwxrwxr-x  2 collins collins 4096 Apr 18  2020 Applications
drwxr-xr-x  2 collins collins 4096 Apr 13  2020 bin
drwxrwxr-x  2 collins collins 4096 Apr  3  2020 cores
dr-xr-xr-x  2 collins collins 4096 Apr 16  2020 dev
drwxrwxr-x  2 collins collins 4096 Apr  3  2020 Developer
lrwxrwxrwx  1 collins collins   11 Apr  3  2020 etc → private/etc
drwxrwxr-x  2 collins collins 4096 Apr 13  2020 Library
drwxr-xr-x  2 collins collins 4096 Apr 19  2020 private
drwxr-xr-x  2 collins collins 4096 Apr 13  2020 sbin
drwxr-xr-x  3 collins collins 4096 Apr  3  2020 System
lrwxrwxrwx  1 collins collins   15 Apr  3  2020 tmp → private/var/tmp
drwxr-xr-x 10 collins collins 4096 Apr 13  2020 usr
lrwxrwxrwx  1 collins collins   11 Apr  3  2020 var → private/var
```

```
┌──(collins㉿kali)-[~/…/JOSH/NoTar-13-4-1/System/Library]
└─$ ls
AccessibilityBundles      ControlCenter            fps                      PreferenceBundles            SyncBundles
AccessoryUpdaterBundles   CoreAccessories          Frameworks               PreferenceManifests          SystemConfiguration
Accounts                  CoreAS                   Health                   PreferenceManifestsInternal  Templates
ApplePTP                  CoreDuet                 HIDPlugins               Preferences                  TextInput
AppleUSBDevice            CoreImage                IdentityServices         PreferencesSyncBundles       ThermalMonitor
AppPlaceholders           CoreServices             KerberosPlugins          PreinstalledAssets           Trial
AppRemovalServices        DataClassMigrators       KeyboardLayouts          PreinstalledAssetsV2         TTSPlugins
AppSignatures             DefaultsConfigurations   LaunchDaemons            PrivateFrameworks            TVAppBundles
AssetTypeDescriptors      DeviceOMatic             LinguisticData           ProceduralWallpaper          UsageBundles
Assistant                 DifferentialPrivacy      LocationBundles          ProductDocuments             UserEventPlugins
Audio                     DistributedEvaluation    Lockdown                 Recents                      UserManagement
AWD                       DoNotDisturb             MediaCapture             RelevanceEngine              UserNotifications
Backup                    Duet                     MediaStreamPlugins       RunningBoard                 VideoCodecs
BridgeManifests           DuetActivityScheduler    Messages                 Sandbox                      VideoDecoders
BulletinDistributor       DuetExpertCenter         NanoLaunchDaemons        ScreenReader                 VideoEncoders
CacheDelete               DuetKnowledgeBase        NanoPreferenceBundles    Security                     VideoProcessors
Caches                    Extensions               NanoTimeKit              SESStorage                   VoiceServices
CardKit                   FDR                      NetworkServiceProxy      SetupAssistantBundles
CardServices              FeatureFlags             Obliteration             SoftwareUpdateCertificates
'Carrier Bundles'         Filesystems              OnBoardingBundles        Spotlight
ColorSync                 Fonts                    PairedSyncServices       SpringBoardPlugins
```